

Learning Whom to Test: Per-Case Credit Assignment for Laboratory Test Allocation in Arbovirus Surveillance

Zuïlho R. C. Segundo Iasmim F. de Almeida Flávio C. Coelho

School of Applied Mathematics, Fundação Getúlio Vargas (FGV EMap)
Rio de Janeiro, Brazil

Abstract

When dengue and chikungunya co-circulate, clinical triage cannot reliably tell them apart, and laboratory confirmation is too scarce to be applied to every suspected case. Deciding *which* suspects to test is a sequential decision made in epidemic time: tests take days to return, results change what should be done next, and hundreds of patients are open at once. We cast this problem as a Markov decision process in a simulator grounded in surveillance data—SEIR dynamics with literature parameters, spatial case distributions estimated by kriging from notifications in Rio de Janeiro and Recife, noisy clinicians, delayed laboratory results, and suspects that have neither disease. In real data, the location of a case barely reveals its disease: a Bayes classifier using location alone is right 52–57% of the time in every sizeable epidemic year of two cities (95% bootstrap intervals within ± 0.01), so tests are the main source of information. Standard deep reinforcement learning fails to learn to use them, because the value of a test is diluted across interleaved decisions about other patients (correlation between the standard return and the test’s outcome: -0.02 ; along the patient’s own timeline: 0.99). We propose a per-case variant of proximal policy optimization that computes advantages along each patient’s timeline, discounted in epidemic days, without changing the environment or the reward. Across four training geographies and four test geographies (five training seeds, 30 paired outbreaks per test), per-case credit outperforms standard advantage estimation in every one of the 16 combinations. Trained on Rio de Janeiro, the agent matches the best expert rule we wrote a priori—sequential testing guided by the clinician’s suspicion—in reward ($+19$, 95% CI $[-89, +121]$) and accuracy, with 7% fewer tests, and stays within 8% of it on two held-out epidemic years of another city. A case-by-case analysis shows what it learned: it follows the clinician’s suspicion for the first test and always orders the second after a negative, exactly like the rule, and saves tests by trusting the clinician when the suspect is labeled non-arboviral. Written down as a rule, that behavior is as good as the expert rule with fewer tests, and better when tests are expensive. The conclusions hold when test cost, test sensitivity and clinician accuracy vary, and on a second set of fresh outbreaks never used during development. When location is informative (an idealized geography), the agent surpasses the rule ($+627$ [$+499, +752$]) with 62% fewer tests, but that policy does not transfer to real cities, and training on a mixture of geographies does not recover either advantage. An agent trained on Recife first failed to transfer; input ablations traced the failure to a spatial shortcut created by a simulator artifact—non-arboviral suspects placed outside the city—and once the artifact is removed, agents trained on either city tie the rule in both.

Keywords: reinforcement learning; credit assignment; arboviruses; diagnostic testing; resource allocation; transfer; epidemiological surveillance.

1 Introduction

Arboviral diseases are among the most persistent public-health burdens in tropical countries such as Brazil [1]. Dengue, chikungunya and Zika share the *Aedes aegypti* vector and cause large

cyclical epidemics that periodically overwhelm health services, with costs that range from lost productivity to hospital and emergency care [2, 3]. When these viruses co-circulate, their clinical pictures overlap, which hampers real-time surveillance [4].

The case definition used by the Brazilian Ministry of Health—sudden fever with arthralgia, myalgia or rash—is sensitive enough to capture most infections, but has low specificity for the viral etiology [5]. Distinguishing dengue from chikungunya from history and physical examination alone is error-prone [6], and a sizeable fraction of suspects turns out to have neither disease. Laboratory confirmation resolves the ambiguity, but during fast-growing outbreaks testing every suspect is financially and logistically infeasible, so managers must ration tests. In practice this rationing is often reactive: tests are spent on clinically obvious cases while atypical or epidemiologically strategic ones remain unconfirmed [4].

Allocating laboratory tests is therefore a *sequential decision problem under uncertainty*. Testing a suspect, relying on epidemiological evidence, waiting for more information, or assigning a final diagnosis consumes finite resources and shapes the quality of future surveillance [7, 8]. Because the true state of the epidemic is only partially observed—through reporting delays, clinical noise and asymptomatic infection—good policies must reason from incomplete information and update decisions over time [8, 9]. Reinforcement learning (RL) is a natural framework for such policies, and simulation environments with a standard interface make it possible to train and compare them [10, 11].

We deliberately keep the decisions inside *epidemic time*: suspects arrive day by day, tests return after a laboratory delay, and a case comes back to the decision-maker when its result is available. This choice is what makes the problem realistic, and it is also what makes it hard to learn. With hundreds of cases open at once, the consequence of a decision about one patient arrives interleaved with the consequences of dozens of decisions about others. Standard RL estimators chain consecutive steps of the trajectory, so the value of ordering a test must travel through many unrelated decisions before it is credited—and in our setting it does not survive the trip.

Contributions.

1. **A data-grounded simulator of test allocation in epidemic time.** Dengue and chikungunya outbreaks follow SEIR dynamics with literature parameters; case locations are drawn from kriging surfaces estimated from notifications in Rio de Janeiro and Recife; suspects include non-arboviral febrile illness; clinicians are noisy; laboratory results return after five days and send the case back to the agent (Section 3).
2. **Evidence, with uncertainty, that location is weakly informative in real data.** A spatial bootstrap of the notifications shows that location predicts the disease 52–57% of the time in every sizeable epidemic year of both cities, far from the 94% of idealized generators (Section 6.1).
3. **A diagnosis of why standard deep RL fails, and a fix.** The standard return is uncorrelated with the outcome of a test; a return along the patient’s own timeline is almost perfectly correlated. Per-case PPO, which estimates advantages along each patient’s timeline in epidemic days, makes the agent investigate without changing the environment or the reward (Sections 4.3 and 6.2).
4. **What the agent learned, as a rule.** Logging every decision shows that in real cities the learned policy is the clinician-guided sequential rule plus one refinement—no test when the clinician labels the suspect non-arboviral. The refinement, written as a fixed rule, is as good as the expert rule with fewer tests and better when tests are expensive; learning discovered it without being told how reliable that label is (Section 6.4).

5. **A cross-geography evaluation.** Agents trained in four geographies are tested in four, with paired outbreaks and a hierarchical bootstrap over training seeds and outbreaks. Per-case credit wins everywhere; the learned policy matches the best expert rule in real cities with fewer tests, surpasses it where geography is informative, and transfers between Rio de Janeiro and held-out years of Recife in both directions—but not from idealized to real geography. A transfer failure that looked like poor learning is traced, by input ablations, to an artifact of the simulator. The conclusions are robust to the simulated test cost, test sensitivity and clinician accuracy (Sections 6.3, 6.5 and 6.6).

2 Related work

Learning whom to test. The closest deployed system is Eva [12], which allocated Greece’s limited COVID-19 tests among arriving travelers in 2020 from their declared characteristics and from the delayed results of earlier travelers. Eva faces the same scarcity and the same reporting delay, but each traveler is tested at most once and the value of a test is its detection yield plus what it teaches about a traveler type. In our problem a patient may receive zero, one or two tests of different kinds, the clinical label is an unreliable prior that the agent must learn to weigh, and a positive result also feeds the epidemiological confirmation of neighbors. Cost-sensitive diagnosis with RL—sequentially acquiring symptoms, features or laboratory panels before a classification [7, 13, 14]—treats one patient per episode: the decision process of a patient is short and self-contained, and credit assignment is not the bottleneck. Our setting interleaves hundreds of such processes in one shared timeline, and that interleaving is what breaks standard advantage estimation. Classical decision analysis frames the same choice through thresholds: test when the probability of disease lies between a no-test and a treat threshold [15]. The sequential expert rule we compare against is a policy of this kind with fixed thresholds; the learned agent can be read as learning thresholds that depend on context.

Reinforcement learning for epidemic response. RL has been used to choose population-level interventions—school closures, mitigation schedules, dynamic health policies—in epidemic simulators [8, 16]. There the action is a policy for a whole population and the credit problem is long delays. Here the actions are individual diagnostic decisions taken in epidemic time, and the credit problem is interleaving.

Credit assignment. Per-case advantage estimation is related to three lines of work. In cooperative multi-agent RL, a shared reward is decomposed across agents, either by learning an additive or monotonic factorization of the value [17, 18] or by a counterfactual baseline that isolates one agent’s contribution [19]. Treating each patient as an agent that shares the world with the others makes our problem an instance of this setting, with one simplification: the environment knows which part of the reward belongs to which case, so the decomposition is exact rather than learned. Action-dependent factorized baselines [20] exploit conditional independence between action dimensions to reduce the variance of policy gradients; our per-case timelines exploit the conditional independence between patients’ outcomes. Methods for delayed rewards redistribute the return along a single trajectory [21, 22]; in our problem the delay itself is short (five days), and what hides the consequence of a decision is the dozens of unrelated decisions in between. Finally, discounting by elapsed time rather than by steps is the standard semi-Markov formulation [23]; we apply it along each patient’s timeline.

Generalization and shortcuts. Deep RL agents overfit to the environments they are trained on [24, 25], and deep networks in general learn shortcuts—features that predict well in training but not under the intended deployment [26]. Randomizing nuisance properties of the simulator during training is the standard remedy [27]. Our cross-geography evaluation is an instance of

this question with variation that comes from real surveillance data rather than from procedural generation, and it shows both a shortcut and an environment artifact that created it (Section 6.5).

Spatial surveillance. We estimate the spatial distribution of each disease by ordinary kriging of aggregated notifications, a standard geostatistical tool for disease mapping [28]; dengue cases cluster in space and time at scales of weeks and hundreds of meters [29]. Our use of these surfaces is modest: they are a generator of plausible case locations and a measure of how much location reveals about the disease, not an inferential target.

3 Simulation environment

The environment implements the Gymnasium interface [10, 30]. Each episode simulates 300 days of concurrent dengue and chikungunya outbreaks in one city. Every parameter reported below is set in version-controlled configuration files, and the environment’s behavior is locked by an automated test suite.

3.1 Epidemic dynamics

Each disease follows a frequency-dependent SEIR model in a closed population [31, 32]:

$$\dot{S} = -\beta \frac{SI}{N}, \quad \dot{E} = \beta \frac{SI}{N} - \sigma E, \quad \dot{I} = \sigma E - \gamma I, \quad \dot{R} = \gamma I, \quad (1)$$

with $\sigma = 1/\text{latency}$, $\gamma = 1/\text{infectious period}$ and $\beta = R_0\gamma$. Notifications follow the flow σE (onset of the infectious phase, which we take as symptom onset); the cumulative curve is integrated numerically and daily new cases are its rounded increments.

Because both viruses pass through the mosquito, the interval between successive cases includes the extrinsic incubation. Rather than a full host–vector model, we use an *effective* human latent period that absorbs both incubations, so that the model’s generation time (latency plus infectious period) matches the serial interval observed in the field (Table 1). For dengue, the intrinsic incubation averages 5.9 days [33], human infectiousness lasts about five days after symptom onset [34], and space–time clustering is strongest at 15–17 days [29]; R_0 in Rio de Janeiro was estimated at 1.70 (2002) and 1.25 (2012) [35]. For chikungunya in Rio, $R_0 = 1.56$ (95% CI 1.46–1.67) assuming a 14-day generation time [36]; only the sum of latency and infectious period is anchored by data. Each episode draws R_0 uniformly within these ranges, with a population of 300 per disease and 1% initially infected. We validate the generator against properties that can be checked against theory—the growth rate implied by R_0 and the final-size equation $z = 1 - e^{-R_0 z}$ (Figure 1)—rather than against the shape of the curves, which can look plausible when the epidemiology is wrong.

Disease	Effective latency	Infectious period	Generation time	R_0
Dengue	11 d	5 d	16 d	1.25–1.70
Chikungunya	8 d	6 d	14 d	1.46–1.67

Table 1: SEIR parameters. R_0 is drawn uniformly within the range at each episode. Sources: [29, 33–36].

3.2 Where cases occur: spatial surfaces from surveillance data

Cases are placed on a 400×400 grid by an interchangeable spatial generator. The default generator uses surfaces estimated by ordinary kriging from georeferenced notifications: counts

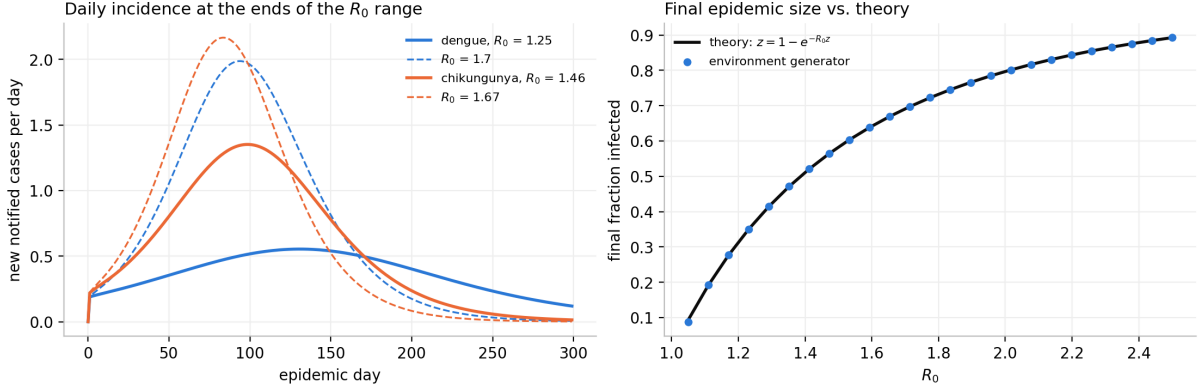


Figure 1: Epidemic dynamics. Left: daily incidence at the ends of the R_0 range for each disease; epidemics last several months. Right: final fraction infected produced by the generator against the final-size equation.

are aggregated in 500 m cells, the log-counts are interpolated, and the result is normalized into a spatial probability per disease, $P(\text{cell} \mid \text{disease})$, from which each day’s cases are sampled. At every episode one of four rigid transformations (identity, 180° rotation, horizontal or vertical mirror) is applied *jointly* to both surfaces, so the agent cannot memorize one city’s geography while the relative geometry of the two diseases—and hence the difficulty—is preserved.

Rio de Janeiro. The reference surfaces use dengue and chikungunya notifications of Rio de Janeiro in 2015–2016 with residence coordinates. Chikungunya barely circulated in Rio in 2015 (70 notifications, against about 14 thousand in 2016), so alternative Rio scenarios combine dengue from 2015 or 2016 with chikungunya from 2016.

Recife. As a second city we use the open notification microdata of Recife [37], which cover dengue (2013–2025) and chikungunya (2015–2025). Each record carries the final classification (including discarded cases), the confirmation criterion, laboratory results and the residence street and neighborhood, but no coordinates. We geocode each record against the city’s official neighborhood polygons and street segments [38]: the street name is matched (after normalizing abbreviations) to the streets that cross the reported neighborhood, or a neighboring one, and a point is drawn along the matched street; when the street is not found, a point is drawn uniformly within the neighborhood. Of 144,175 confirmed or discarded notifications of Recife residents, 72% were placed on the reported street within the reported neighborhood, 6% on the same street in an adjacent neighborhood, 22% at neighborhood level, and fewer than 1% could not be placed. We build one surface per year (2015–2025) from cases confirmed by any criterion, and set the probability to zero outside the municipality, which also removes the sea.

Spatial bootstrap. To quantify the uncertainty of the surfaces and to give training more spatial variety, we build 20 bootstrap replicates for each city and year: the notifications of each disease are resampled with replacement and the kriging is recomputed; in Recife, each replicate is also re-geocoded with a different seed, so the uncertainty of street- and neighborhood-level placement is included. Chikungunya notifications of 2015–2016 coded with the legacy dengue classification scale (4,888 records without any chikungunya-specific result) are excluded as ambiguous.

How informative is location? What matters for the decision problem is not whether each disease has sharp foci, but whether the two diseases occupy *different* places. We summarize this

with the accuracy of the Bayes classifier that sees only the location of a case, with equal priors: $\frac{1}{2} \sum_{\text{cells}} \max\{P(\text{cell} \mid \text{dengue}), P(\text{cell} \mid \text{chik})\}$. It equals 0.5 when location carries no information and 1 when it determines the disease. In Rio 2015–2016 it is 0.544: each surface has clear foci, but the foci coincide (correlation 0.85 between the two surfaces; in 90% of the cells one disease is at most 1.4 times more likely than the other), because both follow where people live and where the vector is (Figure 2, top). The idealized generator used in earlier work—two truncated Gaussian foci—reaches 0.937.

Controlling spatial informativeness. To study sensitivity to this quantity we sharpen the surfaces with a temperature τ ($p \propto p^\tau$; $\tau = 0$ is uniform), which moves the Bayes accuracy from 0.50 ($\tau = 0$) through 0.544 ($\tau = 1$) to 0.94 ($\tau = 32$). Sharpening reaches high separability only by concentrating each disease: half of the dengue cases fall in 34% of the cells at $\tau = 1$, in 1% at $\tau = 8$ and in 8 of 13,195 cells at $\tau = 32$ (Figure 2, bottom). Intermediate values preserve the shape of the foci; $\tau \geq 16$ is an artificial limit rather than a plausible epidemic.

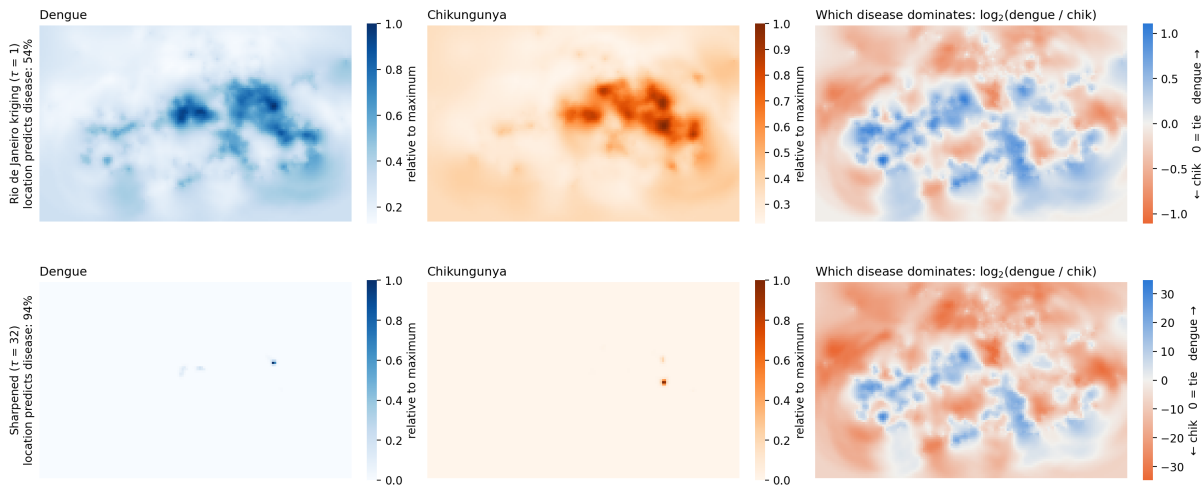


Figure 2: Spatial surfaces of dengue and chikungunya, each on its own scale, and the log-ratio between them. Top: kriging on Rio de Janeiro notifications (2015–2016); each disease has sharp foci, but the ratio stays between $0.5\times$ and $2\times$ and location predicts the disease 54% of the time. Bottom: sharpened surfaces ($\tau = 32$); location becomes highly informative only because each disease collapses onto a handful of cells.

3.3 Case mix: suspects that are neither disease

In real triage, many suspects have another febrile illness. The simulator adds non-arboviral suspects at 25% of the notified cases, uniform in space (they do not follow the vector’s foci) and proportional in time to the arboviral cases of each day. In the main experiments they are uniform over the whole grid; Section 6.5 shows that this becomes an artifact when a city does not fill the grid, and a corrected variant places them uniformly over the inhabited area only. This proportion is consistent with the Recife data: in the epidemic years (2015, 2016 and 2021), discarded notifications are 18–30% of the total. These cases give depth to the investigation: a negative dengue test no longer implies chikungunya, a second test becomes informative, and concluding “not an arbovirus” becomes a legitimate decision.

Two-disease variant. An earlier version of our kriging generator omitted the non-arboviral suspects, so every suspect had one of the two diseases. We keep this configuration as an ablation of the case mix—the *two-disease variant*—because it isolates how the case mix changes which

policy is optimal (Section 6.3). Several robustness analyses were run in this variant and are reported as such.

3.4 Clinicians

The true disease is hidden from the agent and replaced by a noisy clinical label. Each episode draws a clinician whose accuracy is uniform in $[0.50, 0.95]$ and applies to both diseases. Non-arboviral suspects are recognized as such at triage with probability 0.15; otherwise the clinician labels them dengue or chikungunya with equal probability—if they were easy to recognize, they would not be in the queue of suspects. The environment also supports Beta-distributed sensitivity and specificity with controllable mean and dispersion. The clinician turns out to be the dominant source of variance between episodes (Section 5).

3.5 Laboratory tests and epidemiological confirmation

Laboratory tests follow RT-PCR as the diagnostic gold standard: sensitivity 0.95, specificity 0.98 and 2% indeterminate results, which leave the working diagnosis unchanged. These values follow the clinical evaluation of the CDC dengue RT-PCR assay, which detected viral RNA in 98% of acute samples of confirmed cases and gave no false positives in other febrile illnesses [39]; multiplex assays agree less on chikungunya samples with low viral load [40], so Section 6.6 varies the sensitivity from 0.80 to 0.99. Results become available five days after collection. Epidemiological confirmation simulates a field investigation: a case is confirmed if, within a radius of 20 grid cells, there is more than one other case already confirmed *by tests the agent ordered*. It therefore uses only information the surveillance system has, and it makes each positive test an investment in future confirmations nearby. Under a policy that tests every case, it confirms 19.3% of cases with 80.1% precision, against 57.1% accuracy of the clinical label.

3.6 Decision process

The epidemic advances in days; within each day, the cases awaiting a decision are presented to the agent one at a time, so each agent step is a decision about one case (Figure 3). The agent observes a six-channel spatial tensor summarizing notified cases, tests and confirmed cases; the coordinates of the current case; and a 16-dimensional context vector with the clinical label, the results of tests already done, epidemiological confirmation, the local density of confirmed cases, and a two-number summary of the evidence accumulated about the clinician (agreement between test results and clinical labels so far, and how much evidence supports it). This summary tracks the clinician’s true accuracy with correlation 0.99; estimating it requires spending some tests early, which creates a natural exploration–exploitation trade-off.

Actions. For each case the agent chooses among seven actions: **no action**, which costs nothing and lets the clinician’s diagnosis stand; **investigate**, by ordering a dengue test, a chikungunya test, or epidemiological confirmation; and **conclude**, by asserting dengue, chikungunya or “other”. The asserted class comes from the action itself, so the agent can disagree with the clinician without ordering a test.

Case lifecycle. When the agent investigates, the sample is collected immediately but the result matures five days later; **when the result arrives, the case returns to the agent**, who decides with the result in hand. A case leaves the queue when it is concluded, when the agent takes no action on it, or when the episode ends; it can return at most twice, after which only concluding actions are allowed. Without returns, ordering a test would mean giving up the decision about that case, and testing and deciding would be mutually exclusive.

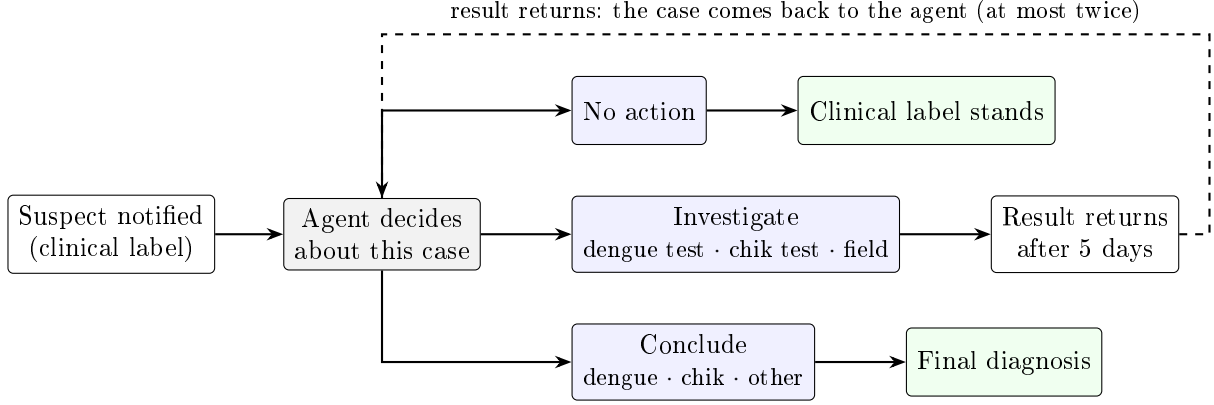


Figure 3: Lifecycle of a suspected case. Each agent step is a decision about one case; hundreds of cases are open at the same time, so the steps of one case are interleaved with those of many others.

3.7 Reward

The reward has three layers. **Immediate cost:** each test or field investigation costs 4.0; no action and concluding cost nothing. **Decision outcome:** a concluding action earns +10 if correct, −30 if it discards a real arbovirus case (a surveillance false negative), and −20 for other errors; concluding an investigated case earns a +2 bonus. **Terminal score:** at the end of the episode each case contributes +1 if its final diagnosis is correct and −3 otherwise, whether or not it was tested, so a test is rewarded for the information it brings and not for being ordered. A case whose investigation was opened and never concluded costs −10, charged when it is abandoned; taking no action on an uninvestigated case is not abandonment but acceptance of the clinical diagnosis, judged by the terminal score.

The test cost governs whether an allocation problem exists: with cheap tests, testing everyone is optimal; with expensive ones, the best policy is to do nothing. At 4.0, investigating pays with a narrow margin—the regime in which choosing *which* cases to investigate matters most. Section 6.6 varies the cost from 2 to 8 in the full environment.

Per-case decomposition. At each step, the environment also reports which case the step’s reward belongs to, on which epidemic day the decision was made, and whether the case is finished; each case’s terminal share is attributed to it exactly once. The per-case shares sum exactly to the episode reward, so the metric used to compare agents is unchanged. This decomposition is the raw material for the credit assignment of Section 4.3.

4 Agents

4.1 Fixed policies

Fixed policies establish what can be achieved without learning; without them, a trained agent can look competent merely by beating chance.

- **Clinical only:** never acts; the clinician’s label stands.
- **Confirm-all:** concludes every case with the clinical label, without testing.
- **Test-once:** orders one dengue test per case and concludes with the result.
- **Test-twice:** orders both tests for every case and concludes with both results.

- **Sequential, dengue first:** orders a dengue test; orders the chikungunya test only if the first is not positive; concludes.
- **Sequential, clinician-guided:** as above, but first tests for the disease the clinician suspects (dengue if the clinician suspects neither).

The sequential rules are the natural expert policies between test-once and test-twice: a positive result settles the case, a negative or indeterminate one calls for the other test. A seventh rule was written *after* the case-by-case analysis of the learned agent (Section 6.4) and is reported separately, since it was not designed a priori:

- **Sequential, trusting “not arboviral”:** as the clinician-guided rule, but a suspect that the clinician labels non-arboviral is concluded as such without any test.

4.2 Learning agents

Deep Q-network. We first trained a DQN [41] with a convolutional encoder. It is kept as a control: once stabilized with a Huber loss, it converges reliably to a degenerate policy of inaction (Section 6.2).

Proximal policy optimization. PPO [42] optimizes a stochastic policy π_θ with the clipped objective

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad (2)$$

where the advantage $\hat{A}_t$ is usually obtained by generalized advantage estimation (GAE) [43] over *consecutive* steps of the trajectory:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad \hat{A}_t = \sum_{k \geq 0} (\gamma \lambda)^k \delta_{t+k}. \quad (3)$$

4.3 Per-case credit assignment

The problem. Consecutive steps of the trajectory belong to different patients. Between a test and the decision it informs there are about 30 decisions about *other* cases; the test itself only returns -4 , and its benefit appears in a later step that belongs to the same case but is surrounded by unrelated rewards. Equation (3) forces that value through all of them, accumulating their noise. We measured the effect directly: at the step of each test, the correlation between the return and whether the test led to a correct diagnosis is -0.019 for the standard return discounted per step, and 0.994 for the return of the case itself discounted per epidemic day. The signal is in the trajectory; the standard definition of the return erases it.

This also explains why three earlier attempts failed. Potential-based reward shaping [44] does not apply: the next state is already another patient, so the potential cancels within the step, and with hundreds of open investigations the drift $(1 - \gamma)\Phi$ exceeds the intended increment for any scale. Retroactive credit would require rewriting rewards already delivered. Making tests cheaper doubled the margin of investigating and did not make the agent order a single test. The obvious alternative—one episode per case—would remove the interleaving by destroying the epidemic dynamics, which is the object of this work.

The method. Each transition carries the episode and case identifiers, the epidemic day d of the decision and the case’s share r^{case} of the reward. *Per-case PPO* groups the transitions of a batch by (episode, case) and computes GAE along each patient’s timeline, with discounting

raised to the number of *days* between decisions rather than steps. For the successive decisions j of one case:

$$\begin{aligned}\text{disc}_j &= \gamma^{d_{j+1}-d_j}, \\ \delta_j &= r_j^{\text{case}} + \text{disc}_j V(s_{j+1}) - V(s_j), \\ \hat{A}_j &= \delta_j + \text{disc}_j \lambda \hat{A}_{j+1}.\end{aligned}\tag{4}$$

At the last step of a case in the batch there is no future if the case is finished; if the batch cut its timeline, we bootstrap from its own value with one day of discount. The structure mirrors credit assignment in cooperative multi-agent systems [17, 19], with each case as an agent sharing the world with the others, and the discount in days is the semi-Markov formulation [23] (Section 2). Two invariances are verified by automated tests: the per-case shares reconstruct the episode reward, so the evaluation metric is unchanged; and the advantage of a case does not depend on the interleaving—inserting 30 noisy decisions about other cases between two steps of case X leaves X ’s advantage unchanged.

Limitation. Per-case credit ignores the externality of a test: a positive result improves epidemiological confirmation for *future* cases nearby, and that benefit does not flow back to the case that paid for the test. The agents barely use epidemiological confirmation, so the effect is currently small.

4.4 Architecture and training

All deep agents share the same network: convolutional layers process the spatial tensor and end in adaptive pooling to $64 \times 6 \times 6$, and a dense branch processes the case coordinates and context vector [45]. Since the pooling discards resolution, the map is delivered at 100×100 , which reduced observations sixteen-fold at no measurable cost. We use Tianshou [46] with 300,000 steps per run, learning rate 3×10^{-4} , $\gamma = 0.99$, $\lambda = 0.95$, $\epsilon = 0.2$, entropy coefficient 0.02, batches of 256 and eight parallel training environments. The action mask that forces a conclusion after two returns is applied to the actor’s logits—otherwise PPO silently ignores it. Rewards are scaled by 0.10 during training only.

To separate the effect of credit assignment from that of temporal information, we trained three arms with the same network and hyperparameters (Table 2): $A \times C$ isolates credit assignment with identical observations; $B \times C$ isolates four optional temporal features (fraction of the horizon elapsed, cases notified today, 7-day trend, age of the case), all computed from what surveillance observes.

Arm	Advantage	Context	Training seeds
A	standard GAE (Eq. 3)	16	4 (two-disease); 3 per geography (full)
B	per case, in days (Eq. 4)	16 + 4 temporal	3 (two-disease)
C	per case, in days (Eq. 4)	16 (as A)	4 (two-disease); 3 per geography (full)

Table 2: PPO arms.

5 Evaluation protocol

Paired outbreaks. Agents are evaluated on 10 test outbreaks, each identified by a fixed seed that reproduces the same epidemic, the same clinician and the same clinical noise for every agent, so comparisons are *paired*. This matters: the clinician drawn in an episode explains most of the variance—its accuracy correlates at **0.97** with the episode reward of both the clinical policy and

a trained agent. A second set of 12 never-used seeds tests generalization. We always evaluate the *final* checkpoint; selecting the best checkpoint during training selects noise (it scored 1,000–1,460 points above the final checkpoint for DQN).

Hierarchical, paired bootstrap. There are two sources of variance: the seed that trained the agent and the outbreak it is evaluated on. Confidence intervals come from a hierarchical bootstrap [47] that resamples training seeds and then outbreaks; outbreak resampling is *shared* across arms within each replicate, preserving the pairing, while training seeds—independent runs—are resampled separately per arm. Following Agarwal et al. [48], we report the mean, the interquartile mean, the paired difference between arms and the probability of improvement $P(X > Y)$, the chance that an episode of X beats the same episode of Y over all pairs of training seeds (ties count $1/2$), with 95% percentile intervals from 10,000 replicates. With five training seeds the upper level still has limited resolution, so intervals are honest about outbreaks and somewhat optimistic about training seeds.

Metrics. The primary metric is the episode reward; secondary metrics are three-class diagnostic accuracy (dengue, chikungunya, other) and the number of laboratory tests per episode. In the two-disease variant no case is “other”, so accuracy is effectively binary.

Training geographies and cross-geography evaluation. All cross-geography experiments use the full environment (with non-arboviral suspects) and differ only in where cases occur. Arms A and C are trained with five seeds each (45–49) in four geographies, all in the corrected environment of Section 6.5 (non-arboviral suspects only in the inhabited area of each city):

- **Rio de Janeiro:** each episode draws one of the 20 bootstrap replicates of the 2015–2016 surface.
- **Recife:** each episode draws one replicate of one training year (2017–2020 and 2022–2025; 160 surfaces). The two largest epidemics with both diseases, 2016 and 2021, are held out for testing; 2015 is excluded (only 201 confirmed chikungunya cases, and the epicenter of the Zika epidemic).
- **Idealized:** two well-separated Gaussian foci (location predicts the disease 94% of the time).
- **Mixture:** one third of the episodes from each of the three above, trained for twice the budget (600 thousand steps).

Every trained agent and every fixed policy is then evaluated in four test geographies, always on the *original* surfaces and never on training replicates: Rio de Janeiro 2015–2016, Recife 2016 and Recife 2021 (held out from Recife training), and the idealized generator. Each test uses 30 paired outbreaks with seeds never used before (3001–3030).

The correction, and the experiments that led to it. An earlier round (three seeds) placed non-arboviral suspects anywhere on the grid; Section 6.5 shows why that was an artifact. That round also tested two remedies that we report as part of the diagnosis: Recife with the city randomly rotated, rescaled (35–100% of the grid) and repositioned at every episode, and Rio de Janeiro and Recife at a common physical scale (200 m per cell, the city translated to a random position at each episode). The inhabited area of each city is derived from its notifications by the same rule—500 m cells with at least two notifications of any disease in any year, morphologically closed and with holes filled. For Recife this rule recovers the official municipal polygon almost exactly (875 against 879 cells); for Rio de Janeiro it covers 1,160 km², against 1,200 km² for the municipality. In Rio de Janeiro and in the idealized generator the correction changes nothing, because the surface fills the grid or does not exist.

Sensitivity to simulated parameters. The Rio-trained agents (five seeds) and the fixed rules are also evaluated, without retraining, on the Rio test outbreaks with the test cost set to 2, 3, 6 or 8 (4 in training), the RT-PCR sensitivity set to 0.80, 0.90 or 0.99 (0.95 in training), and the clinician accuracy fixed at 0.55, 0.70, 0.85 or 0.95 (drawn from 0.5–0.95 in training). Two further groups of per-case agents (three seeds each) are trained on Rio with test costs of 2 and 8.

Confirmation set. The derived rule of Section 6.4 and the correction of Section 6.5 were designed after looking at results on the test outbreaks 3001–3030. To remove that selection, the full transfer matrix and the expensive-test conditions (costs 6 and 8) are evaluated again on 30 fresh outbreaks (seeds 4001–4030) that played no part in any decision (Section 6.7).

6 Results

6.1 In real data, location barely reveals the disease

Recife offers eleven years of both diseases in one city, with epidemics of very different composition—dengue-dominated in 2015–2016 and 2019, chikungunya-dominated in 2021 (Figure 4, left). In every year, the Bayes accuracy of location alone stays close to Rio’s 0.544: between 0.520 and 0.573 from 2016 to 2025 (Figure 4, right). The highest value, 0.605 in 2015, comes from only 201 confirmed chikungunya cases, which makes that surface artificially concentrated. The places where one disease dominates change from year to year (Figure 5), but the magnitude of the difference stays small. Recife 2016, the largest epidemic with both diseases, is strikingly similar to Rio 2015–2016 (Bayes accuracy 0.557 vs. 0.544; correlation between the two diseases’ surfaces 0.81 vs. 0.85).

The spatial bootstrap confirms that these values are precise: the 95% intervals are 0.541 [0.539, 0.545] for Rio de Janeiro 2015–2016, 0.556 [0.552, 0.559] for Recife 2016 and 0.530 [0.528, 0.535] for Recife 2021, and, apart from 2015 (0.60, driven by only 201 chikungunya cases), no Recife year exceeds 0.57 (all intervals within ± 0.01). The correlation between the two surfaces, by contrast, is unstable in years with few cases (e.g., $[-0.06, 0.23]$ in 2024) and is only interpreted for the large epidemics (Rio 2015–2016: 0.80 [0.77, 0.82]; Recife 2016: 0.80 [0.79, 0.82]; Recife 2021: 0.95 [0.93, 0.96]).

The implication for the decision problem is direct. The idealized generator, with location 94% informative, lets an agent diagnose from the map; real surveillance data do not. In earlier work with that generator, a PPO agent reached a positive reward without ordering a single test, by disagreeing with the clinician based on location. In real geography, the laboratory test is the main source of information, and the allocation problem is real.

6.2 Standard deep RL does not learn to investigate; per-case credit does

DQN and standard PPO. DQN was unstable, with rewards oscillating by up to 6,300 points between neighboring evaluations. A Huber loss stabilized it unambiguously (from ± 1340 to ± 9 across seeds) and made it *worse*: stabilized, it converged reliably to 66.7% inaction and no tests (-4164 ± 9). With the same environment, network and action mask, PPO with standard GAE was the first learned agent to beat doing nothing ($+632 \pm 43$), with monotonic training curves. Learning rate, buffer size, n -step returns and target-update frequency did not move DQN measurably. PPO, however, hardly investigated: 7 to 37 tests per episode in the Rio geography, against about 370 for test-once.

Per-case credit. Table 3 and Figure 6 show the three arms in the two-disease variant. Three conclusions follow.

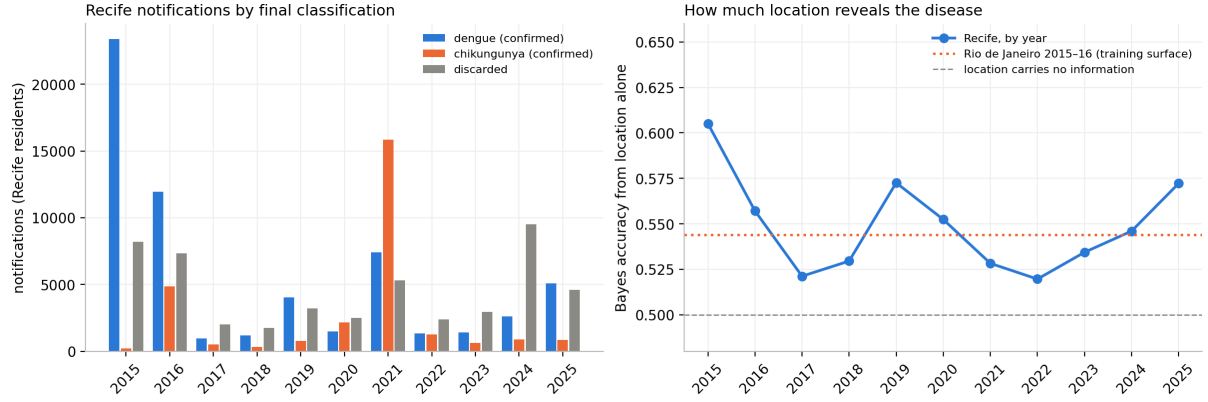


Figure 4: Recife, 2015–2025. Left: notifications of residents by final classification. Right: accuracy of a Bayes classifier that sees only the location of a case, per year, against the Rio de Janeiro surface used for training (dotted) and the no-information level (dashed).

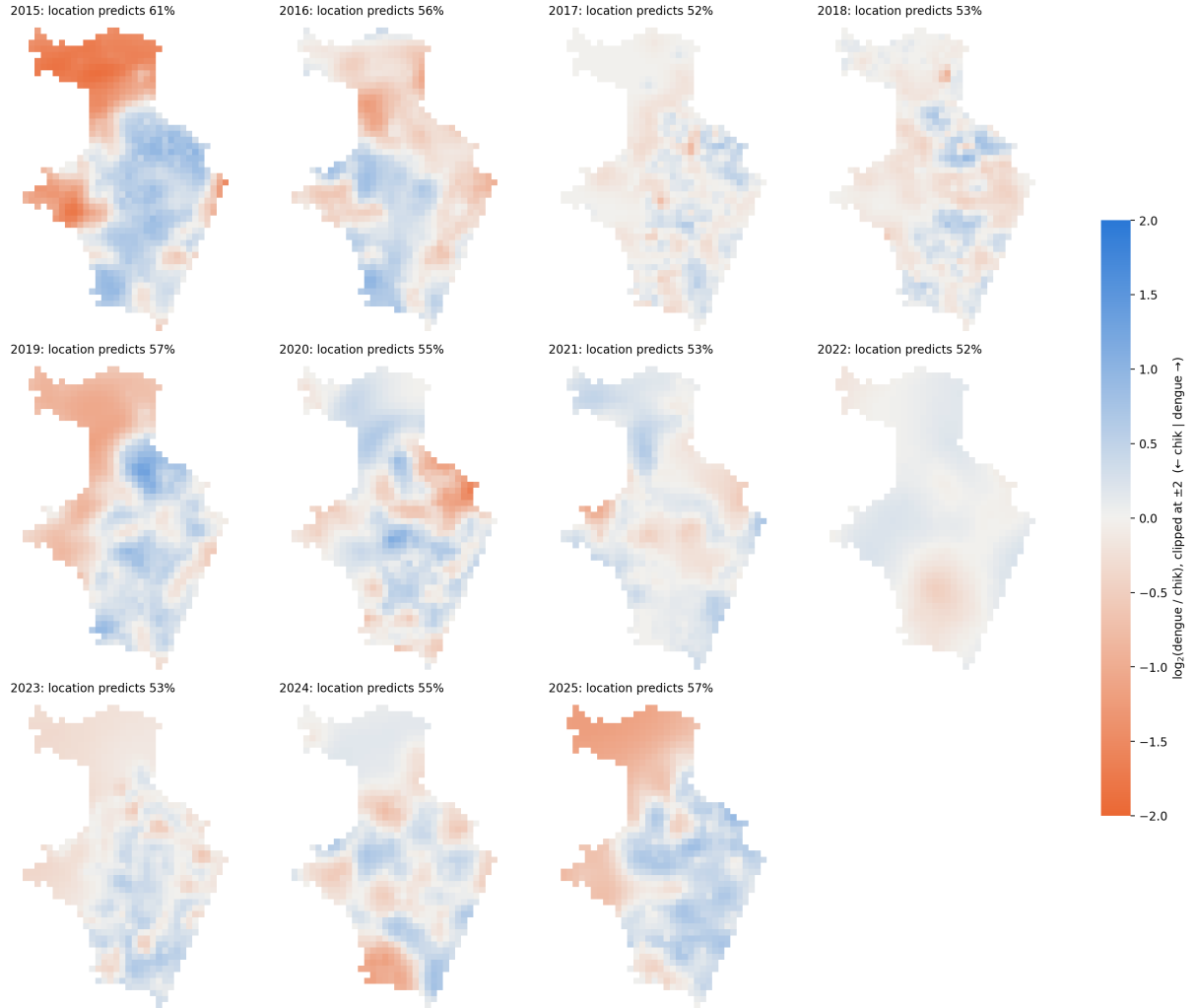


Figure 5: Where each disease dominates in Recife, year by year: $\log_2(P(\text{cell} | \text{dengue})/P(\text{cell} | \text{chik}))$ from kriging surfaces, clipped at ± 2 . The map changes between years, but the contrast remains weak.

The gain comes from credit assignment. With identical observations, C outperforms A by +2009 [+1197, +2882] with $P(C > A) = 1.00$: in every outbreak, for every pair of training seeds, the per-case agent wins. It moves from 25 to 284 tests per episode and gains 20 accuracy points. A also collapses in specific outbreaks (down to -2200 in those with poor clinicians), where it inherits the clinician’s errors because it barely tests; C never collapses.

The temporal features do not contribute. B and C are indistinguishable ($-43 [-196, +85]$, $P = 0.47$). This does not show that temporal information is useless, only that these four features added nothing on top of per-case credit.

The per-case agent matches the best fixed rule with fewer tests. Without non-arboviral suspects, a single test settles a case and test-once is the best fixed policy. C’s reward is statistically indistinguishable from it ($-90 [-250, +67]$) with 84 fewer tests per episode (-23%) and 2.2 fewer accuracy points. On 12 unseen outbreaks the pattern holds: $C \times A = +1653 [+1072, +2313]$, $P = 1.00$; $C \times \text{test-once} = +6 [-124, +149]$. An input ablation without retraining shows that zeroing the entire spatial map changes only 0–2% of the agent’s decisions: the policy decides from the case context, consistent with the weak spatial signal of Section 6.1.

Policy	Seeds	Reward [95% CI]	Accuracy [95% CI]	Tests [95% CI]
Test-once (best fixed rule)	–	+2782 [+2668, +2900]	0.960 [0.953, 0.966]	369 [353, 384]
C: per-case credit	4	+ 2692 [+2523, +2842]	0.937 [0.923, 0.950]	284 [223, 335]
B: per-case credit + time	3	+2649 [+2413, +2842]	0.931 [0.908, 0.946]	266 [206, 321]
Test-twice	–	+1051 [+969, +1135]	0.950 [0.945, 0.954]	737 [706, 768]
A: standard GAE	4	+684 [–146, +1498]	0.739 [0.674, 0.803]	25 [5, 64]
Confirm-all	–	+346 [–656, +1335]	0.710 [0.633, 0.789]	0
Clinical only	–	–64 [–184, +54]	0.710 [0.633, 0.789]	0

Table 3: Two-disease variant (Rio de Janeiro surfaces), 10 paired outbreaks; final checkpoints; hierarchical, paired bootstrap. Paired contrasts: $C \times A = +2009 [+1197, +2882]$, $P = 1.00$; $B \times C = -43 [-196, +85]$, $P = 0.47$; $C \times \text{test-once} = -90 [-250, +67]$, $P = 0.38$.

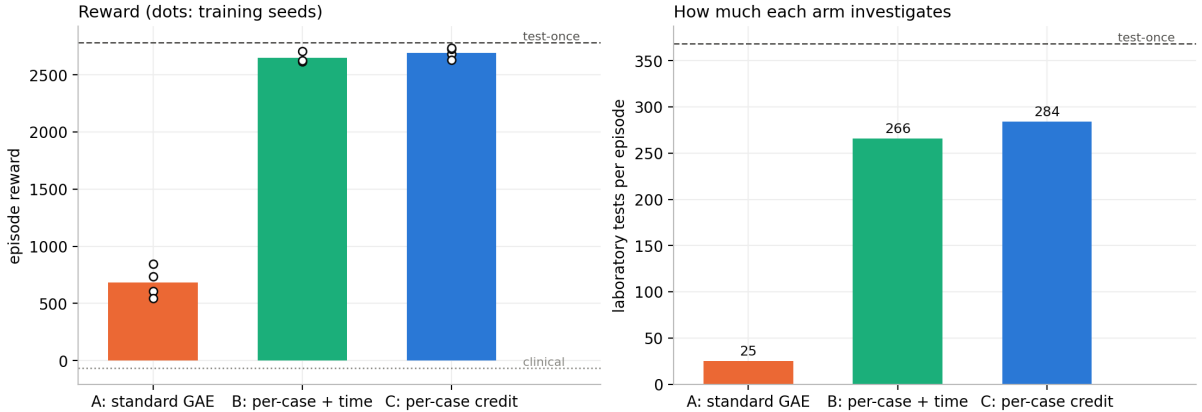


Figure 6: The three PPO arms in the two-disease variant. Left: mean episode reward (dots are training seeds). Right: laboratory tests per episode. Dashed lines: test-once; dotted: clinical only.

6.3 With non-arboviral suspects: the case mix decides the best rule

Adding non-arboviral suspects changes the problem, not just the numbers (Table 4). With only two diseases, a negative dengue test implies chikungunya and test-once is optimal. With a quarter of the suspects being neither, a negative test is ambiguous: test-once drops from +2782 to +322

and 0.756 accuracy, and investigation must go deeper. The idealized generator, which always had non-arboviral suspects, shows the same ordering as the full environment—the reordering is caused by the case mix, not by geography. The best fixed rule is now sequential testing guided by the clinician: testing first for the suspected disease settles most cases with one test when the clinician is right.

Fixed policy	Kriging, two-disease	Kriging, full	Idealized generator
Test-once	+ 2782	+322	+473
Test-twice	+1051	+1225	+1249
Sequential, clinician-guided	–	+ 2330	–
Confirm-all	+346	–1734	–1821
Clinical only	–64	–344	–355

Table 4: Fixed policies with SEIR dynamics on the 10 benchmark outbreaks. Removing non-arboviral suspects (two-disease variant) is what makes a single test sufficient; the idealized generator, which has them, orders the policies as the full environment does.

The per-case agent matches the best expert rule with fewer tests. Table 5 and Figure 7 show agents trained on Rio de Janeiro, evaluated on the original Rio surface over 30 new paired outbreaks with five training seeds. C reaches +2168 against +2149 for the clinician-guided sequential rule: the difference, +19 [–89, +121], is not distinguishable from zero ($P(C > \text{rule}) = 0.53$), and neither is the difference in accuracy (–0.006 [–0.014, 0.000]). C orders 49 fewer tests per episode [–62, –39], a 7% saving. Against the other fixed rules C is clearly better: about +370 over sequential dengue-first and +1020 over test-twice, with fewer tests than both. The agent was never told about test order or the clinician’s reliability; it learned a policy as good as the best rule we could design, and a more frugal one.

The standard agent collapses. With standard GAE, A orders about 63 tests per episode, its accuracy stays near the clinician’s, and its reward falls well below doing nothing (–1743 against –368). The gap between the arms is +3911 [+3395, +4438], with $P(C > A) = 1.00$.

Policy	Seeds	Reward [95% CI]	Accuracy [95% CI]	Tests [95% CI]
C: per-case credit	5	+ 2168 [+2042, +2290]	0.935 [0.927, 0.940]	651 [621, 679]
Sequential, trusting “not arbovira” [†]	–	+2205 [+2087, +2315]	0.939 [0.935, 0.943]	661 [634, 687]
Sequential, clinician-guided	–	+2149 [+2043, +2254]	0.941 [0.938, 0.945]	699 [669, 728]
Sequential, dengue first	–	+1799 [+1700, +1897]	0.937 [0.933, 0.941]	772 [749, 794]
Test-twice	–	+1144 [+1064, +1228]	0.937 [0.933, 0.941]	931 [899, 963]
Test-once	–	+407 [+323, +488]	0.763 [0.757, 0.768]	466 [449, 481]
Clinical only	–	–368 [–436, –299]	0.551 [0.515, 0.588]	0
A: standard GAE	5	–1743 [–2299, –1193]	0.580 [0.544, 0.616]	63 [27, 103]

Table 5: Full environment, agents trained and evaluated on Rio de Janeiro: 30 paired outbreaks never seen before, five training seeds per arm, hierarchical and paired bootstrap. Paired contrasts: C × clinician-guided sequential = +19 [–89, +121], $P = 0.53$, tests –49 [–62, –39]; C × A = +3911 [+3395, +4438], $P = 1.00$. [†]Rule written after inspecting the learned agent (Section 6.4); C × this rule = –37 [–138, +56].

6.4 What the agent learned

Aggregate reward says that the agent is as good as the rule, not *how*. We logged every decision of the three Rio-trained C agents and of the rule on the same 30 outbreaks—about 35 thousand decisions per agent and test—and summarized them per case (Figure 8).

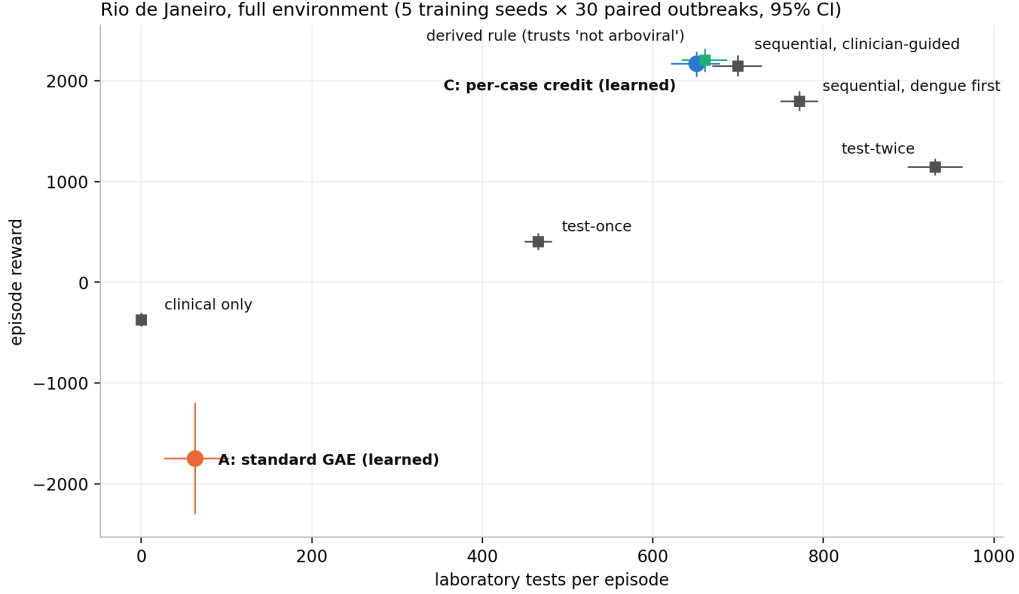


Figure 7: Reward against tests per episode in the full environment on Rio de Janeiro (means with 95% intervals over five training seeds and 30 paired outbreaks). The per-case agent sits next to the best expert rule and the rule derived from it (green), with fewer tests; the standard agent collapses near inaction.

In real cities the agent is the rule plus one refinement. For suspects the clinician labels dengue or chikungunya, the agent orders the first test for the suspected disease in 99.3–99.6% of cases and orders the second test after a negative first result in 99.9–100%—exactly what the clinician-guided rule does. It does *not* choose which negatives deserve a second test; it always orders it—and, as we show next, that is the optimal choice in this environment. The saving comes from somewhere else: of the suspects labeled non-arboviral (4.3% of cases, of which 84% are indeed non-arboviral), the agent concludes 94% as such without testing, where the rule orders 1.9 tests each. Accuracy on those cases falls from 0.96 to 0.85, a price that 1.9 tests per case outweighs at our test cost. In Recife the agent also leaves 8.4% of dengue-labeled suspects untested (2.9% in Rio), which accounts for its lower accuracy there.

Always ordering the second test is optimal here. After a negative first result, concluding immediately with class c is worth $10p_c + e_c(1 - p_c)$, where p_c is the posterior probability of c and e_c is the penalty for a wrong conclusion (-30 for “not arboviral”, the missed case; -20 otherwise). Ordering the second test is worth what the rule actually obtains after it, minus the test’s cost ($+2.2$ per case in Rio de Janeiro). We estimated p_c for each case with a cross-validated multinomial logistic regression on everything the agent observes at that moment: the clinician’s label, which test came back negative, the local density of confirmed cases of each disease and the epidemic day. The posterior is flat: its largest class has a median probability of 0.52 in Rio de Janeiro (99th percentile 0.70), because after a negative dengue test the remaining candidates—chikungunya and a non-arboviral illness—are about equally likely for almost every case. Concluding immediately would be better for only 0.4% of the first negatives in Rio (0.6% in the corrected Recife environments), with a gain below 3 reward points per such case—less than one point per episode in total. The agent is therefore right not to select: in this environment there is no case-level information that separates the negatives that need a second test from those that do not. Before the correction of Section 6.5, the same calculation in Recife gives 16–17%, all driven by the artifact: suspects outside the city are known to be non-arboviral.

Written down, the refinement is a frugal rule. We wrote that behavior as a fixed rule—clinician-guided sequential testing, with no test for suspects labeled non-arboviral—and evaluated it on the same outbreaks. It is as good as the clinician-guided rule, +56 [−26, +138], with 38 fewer tests per episode [−43, −34] and the same accuracy (Table 5); the reward difference is not significant at our test cost, but becomes so when tests are more expensive (Section 6.6). The Rio-trained agent ties it in Rio (−37 [−138, +56]) and falls slightly below it on the held-out Recife years (−150 [−315, −18] and −108 [−275, +14]), where it also skips some dengue suspects. The rule, which uses no spatial input, is by construction unaffected by geography. Where location is informative the picture is different: the idealized-geography agent tests only 46% of the cases, follows the clinician’s suspicion for the first test in only 47%, orders a second test after only 46% of negatives, and beats the derived rule by +562 [+431, +685] with about 390 fewer tests.

The derived rule is a direct product of learning: nothing in the rule designer’s prior told us that the non-arboviral label was reliable enough to act on, and the agent found it from reward alone. It also shows the limit of what the agent *could* learn in real cities: with location nearly uninformative and the clinician’s label as the only case-specific prior, the best policy is simple enough to write down. Selecting which cases need a second test requires case-level information that the simulator does not yet provide—symptoms with disease-specific likelihoods, or days since symptom onset, which govern RT-PCR sensitivity—or a laboratory capacity that makes some tests unaffordable.

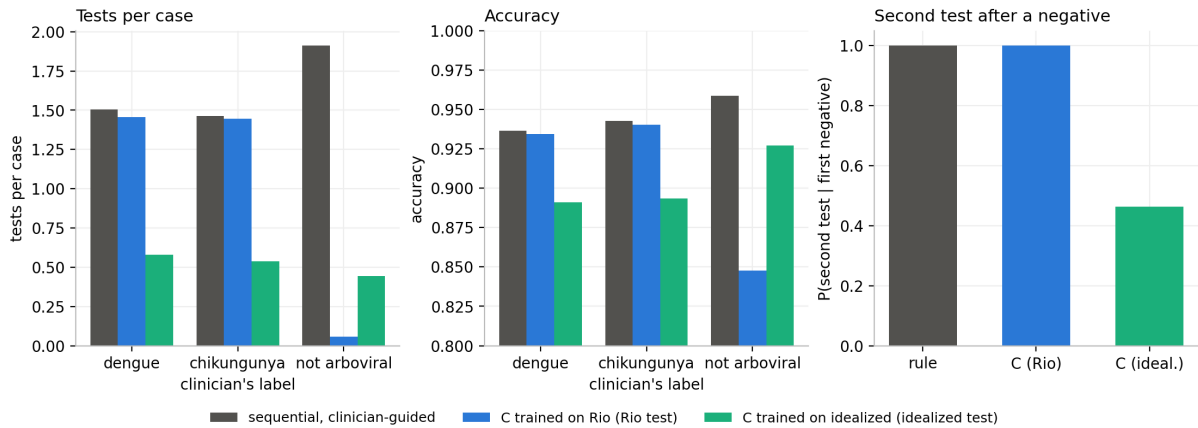


Figure 8: What the per-case agent does, case by case (30 outbreaks; three training seeds for each agent). Left and center: tests per case and accuracy by the clinician’s label. The Rio-trained agent (blue) behaves like the clinician-guided rule (gray) for suspects labeled dengue or chikungunya and concludes suspects labeled non-arboviral without testing. Right: probability of a second test after a negative first result, for suspects labeled dengue or chikungunya. The idealized-geography agent (green), for which location is informative, tests far less and skips most second tests.

6.5 Transfer across geographies

Figure 9 summarizes the cross-geography evaluation in the corrected environment: for each training geography and each test geography, the paired difference between the per-case agent and the clinician-guided sequential rule.

Per-case credit wins everywhere. In all 16 combinations of training and test geography, C outperforms A trained in the same geography, by +1215 to +4275 in reward, with $P(C > A)$ between 0.95 and 1.00. The failure of standard advantage estimation is not specific to one city or geography.

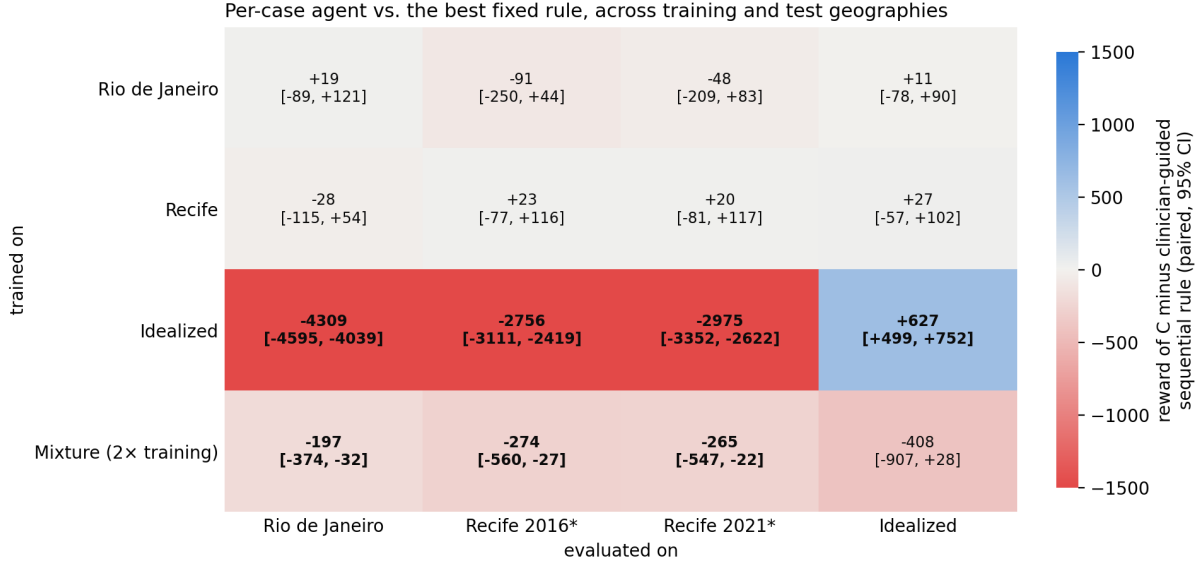


Figure 9: Per-case agent minus the clinician-guided sequential rule, by training geography (rows) and test geography (columns). Cells show the paired difference in episode reward with its 95% interval (five training seeds $\times$ 30 paired outbreaks; corrected environment); bold marks intervals that exclude zero. Recife 2016 and 2021 are held out from Recife training. The rule does not use location, so it is the same yardstick in every column (about +2150).

Policies learned in either real city transfer to the other. The Rio-trained agent stays statistically tied with the rule on the two held-out Recife years (-91 $[-250, +44]$ and -48 $[-209, +83]$; on fresh outbreaks it falls slightly below, Section 6.7), ordering about 62 fewer tests, and on the idealized geography ($+11$ $[-78, +90]$). The Recife-trained agent ties the rule in all four tests, including Rio de Janeiro (-28 $[-115, +54]$), with about 52 fewer tests, and it is the most stable learned policy in the study (seed standard deviation 21 in Rio).

When location is informative, learning beats the rule—but does not transfer. Trained on the idealized geography, the agent surpasses the rule on that geography by $+627$ $[+499, +752]$ while ordering 433 fewer tests (62% fewer), at a cost of 4 accuracy points: it learns to read the disease from the map and saves tests where the map is decisive. The same agent collapses in real cities (-2756 to -4309 below the rule), where location carries almost no information.

A transfer failure traced to a simulator artifact. In the first round, before the correction (three seeds; Figure 10, left), the Recife-trained agent fell below the rule even on its own held-out years (-168 $[-304, -44]$ and -220 $[-366, -73]$) and fails in Rio de Janeiro (-1643 $[-2308, -933]$, seed standard deviation 722). Input ablations without retraining (10 outbreaks per condition) show why. The Rio-trained agent is insensitive to its spatial inputs: zeroing the map, the case coordinates or the local density of confirmed cases leaves its reward between $+2075$ and $+2295$ in both cities. The Recife-trained agent depends on them: without the case coordinates it collapses even in Recife ($+2135 \rightarrow -4631$), and without the map it transfers much *better* to Rio ($+464 \rightarrow +1514$). Rescaling the density features to the training range does not help ($+291$). The cue it uses is an artifact of the simulator: non-arboviral suspects were placed uniformly over the whole grid, but Recife’s arboviral surfaces are zero outside the municipality, which covers only 44% of the grid. A suspect outside the city was therefore certainly non-arboviral—about 14% of all suspects, identifiable from position alone. Rio’s surface fills the grid, so the cue does not exist there. Our first remedy, randomly rotating, rescaling and repositioning the city at every episode, made matters worse (Recife 2016: -430 ; Rio: -3491), consistently with this diagnosis:

shrinking the city enlarges the empty area where only non-arboviral suspects appear.

Removing the artifact restores transfer in both directions. Placing non-arboviral suspects only in the inhabited area, and changing nothing else, gives the corrected environment used in Figure 9: the Recife-trained agent now ties the rule everywhere, with a seed standard deviation of 21 in Rio (722 before the correction).

A common physical scale is not enough by itself. Our environment stretches every city to the same 400×400 grid, so the 20-cell radius of epidemiological confirmation is about 1 km in Recife and 3 km in Rio. Placing both cities at 200 m per cell, in their real extent and at a random position (three seeds; Figure 10, right), makes the Recife agent transfer to Rio with a small loss (-143 [$-268, -42$]) while tying the rule at home, but breaks the Rio agent in Recife (-783 and -835). The reason is measurable: each episode has the same number of cases regardless of city size, so Recife, five times smaller, becomes five times denser. The local density of confirmed cases saturates its normalization in 83% of decisions in Recife, against 37% in Rio at the same scale and 0.3% in the original environment. A physically consistent simulator needs case counts that scale with each city’s population as well as a common spatial scale; with only the scale fixed, it creates a new shift instead of removing one.

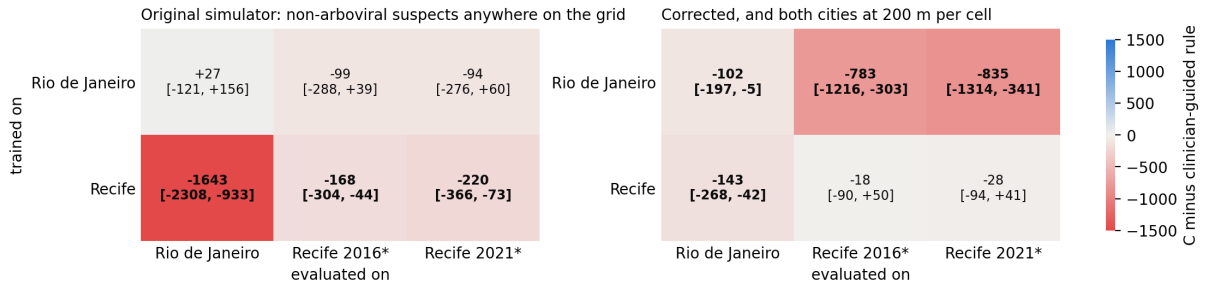


Figure 10: Diagnosing the transfer failure (first round, three training seeds $\times$ 30 outbreaks). Left: the original simulator, with non-arboviral suspects anywhere on the grid; the Recife-trained agent fails even in its own held-out years and collapses in Rio. Right: corrected, and with both cities at 200 m per cell in their real extent; the smaller city becomes denser because each episode has the same number of cases, and the Rio-trained agent fails there. Paired differences with the clinician-guided rule, 95% intervals; bold marks intervals that exclude zero. Compare with the corrected environment in Figure 9.

Scaling the cases with the population removes that shift. We then gave each city a number of cases proportional to its population: the outbreak size of the Rio simulator (300) is kept for Rio de Janeiro, and Recife receives $300 \times 1,488,920/6,211,423 = 72$ (2022 census), about a fifth of the cases. Measured on the generated outbreaks, the density of cases per 1000 inhabited cells goes from 17.2 in Rio and 89.5 in Recife (a factor of 5.2) to 17.2 and 18.5 (a factor of 1.08). With five seeds (45–49) for the Rio agent and for a new Recife agent trained on the scaled outbreaks, and the same 30 paired outbreaks, Table 6 shows two things. First, the Rio-trained agent no longer fails in Recife: at the original density it falls 525 [$-958, -128$] and 576 [$-1031, -176$] below the rule on the two held-out Recife years, whereas on the scaled outbreaks it falls only 38 [$-84, +2$] and 26 [$-70, +11$] below (about 9% and 6% of the rule’s reward of 446), with all five seeds between 376 and 440. The density shift, and not the geography, explained the failure. Second, the Recife agent trained on the scaled outbreaks does not do better at home than the Rio agent (-41 [$-103, +18$] and -49 [$-105, +6$]); it orders fewer tests (71–91 against 132 for the rule) and is less accurate (0.84–0.88 against 0.94), and it does not transfer to larger

outbreaks, where it falls 2330 $[-2679, -2006]$ below the rule in Rio with seed-level rewards from -552 to $+198$. The agent trained on the denser outbreaks (three seeds, previous section) ties the rule in Recife at both densities and loses 143 in Rio, so, in this experiment, training on the larger outbreaks was the more robust choice. We did not test why the agent trained on small outbreaks under-orders tests; one candidate is that the reward per episode is five times smaller while the learning hyperparameters were tuned at the Rio scale.

Table 6: Case counts proportional to the population (common physical scale of 200 m per cell; 30 paired outbreaks; hierarchical, paired bootstrap). Entries are the agent’s reward minus that of the clinician-guided sequential rule, with 95% intervals; the rule’s reward is $+2178$ at the original density (Rio, and Recife with Rio’s number of cases) and $+446$ on the scaled outbreaks. The agent trained on the original density has three seeds; the others, five.

Trained on ↓ / tested on →	Original density		Scaled to population	
	Rio	Recife 2016	Recife 2016	Recife 2021
Rio	$-87 [-168, -11]$	$-525 [-958, -128]$	$-38 [-84, +2]$	$-26 [-70, +11]$
Recife, original density	$-143 [-275, -39]$	$-18 [-91, +50]$	$-22 [-63, +16]$	$-30 [-68, +5]$
Recife, scaled to population	$-2330 [-2679, -2006]$	$-2374 [-2592, -2172]$	$-41 [-103, +18]$	$-49 [-105, +6]$

Mixing geographies does not pay. An agent trained on one third of each geography, with twice the training budget, falls below the rule in all three real tests ($-197 [-374, -32]$, $-274 [-560, -27]$ and $-265 [-547, -22]$) and does not recover the idealized specialist’s advantage there ($-408 [-907, +28]$ against the rule); it also varies more between seeds than any single-geography agent (standard deviation 175–572). In an earlier round with three seeds and the uncorrected Recife environment, the same budget had seemed to tie the rule; with five seeds it does not. A mixture asks one policy to behave in opposite ways—test almost everything in real cities, read the map in the idealized one—and our agents do not learn to tell the regimes apart from the map.

6.6 Sensitivity to simulated parameters

Test cost, test sensitivity and clinician accuracy are parameters of the simulator, not measurements. Figure 11 evaluates the Rio-trained agents, without retraining, as each of them varies.

Test sensitivity and clinician accuracy. With RT-PCR sensitivity between 0.80 and 0.99 and clinician accuracy fixed anywhere between 0.55 and 0.95, the agent stays statistically tied with the clinician-guided rule (every interval contains zero) and keeps ordering about 45 fewer tests. Rewards move a lot with these parameters—from -226 to $+2821$ for the rule as sensitivity goes from 0.80 to 0.99—but the agent and the rule move together.

Test cost. The agent and the derived rule save tests, so their advantage grows with the price of a test: tied with the rule at costs 2–4, they beat it at cost 6 (agent $+116 [+16, +212]$; derived rule $+133 [+47, +219]$) and at cost 8 ($+213 [+116, +309]$ and $+209 [+120, +300]$); on fresh outbreaks only the advantage at cost 8 is reproduced (Section 6.7). An agent retrained at cost 2 also ties the rule there ($-34 [-115, +42]$). Retraining at cost 8 fails: the agent orders only 106 tests, reaches 0.63 accuracy and falls below doing nothing (-1576 against -368). At that cost no policy that tests beats inaction—the regime in which, as noted in Section 3, there is no allocation problem left—and per-case credit does not find the narrow policies that would still pay.

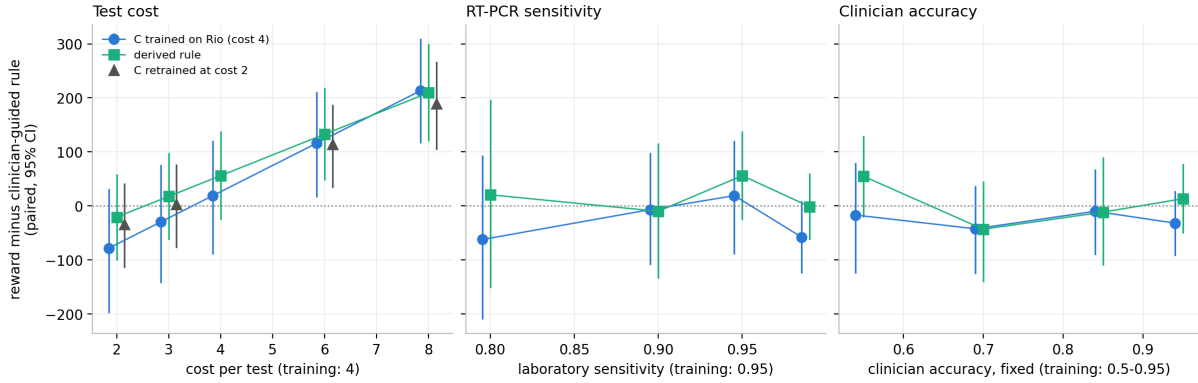


Figure 11: Sensitivity to simulated parameters on the Rio de Janeiro test outbreaks: reward of the Rio-trained per-case agent (blue; trained at cost 4, five seeds), of the derived rule (green) and of an agent retrained at cost 2 (gray), minus the clinician-guided rule, with paired 95% intervals. Left: test cost; center: RT-PCR sensitivity; right: clinician accuracy, fixed for the whole episode.

6.7 Confirmation on fresh outbreaks

On the 30 fresh outbreaks, the main conclusions are reproduced, and two are qualified.

Reproduced. Per-case credit beats standard GAE in all 16 combinations (+1280 to +4142). The Rio-trained agent ties the clinician-guided rule in Rio ($-9 [-109, +89]$) with 49 fewer tests; the Recife-trained agent ties it in all four tests ($-66 [-158, +29]$ in Rio, $-85 [-189, +16]$ and $-104 [-224, 0]$ on the Recife years); the idealized-geography agent beats the rule there (+605 [+501, +702]) and collapses in real cities; the mixture falls below the rule in every real test (-299 to -420). The derived rule is again as good as the clinician-guided rule (+8 $[-90, +109]$ in Rio) with 40 fewer tests, and better at cost 8 (+167 [+61, +276]); so is the Rio-trained agent at cost 8 (+186 [+101, +277]). Retraining at cost 8 fails again.

Qualified. The transfer from Rio de Janeiro to Recife carries a small but detectable cost: on the fresh outbreaks the Rio-trained agent falls slightly below the rule on both Recife years ($-177 [-373, -8]$ and $-150 [-332, -5]$, about 7–8% of the rule’s reward), where on the first set it was tied. The advantage of the derived rule at cost 6 is not reproduced (+88 $[-14, +193]$); it holds only at cost 8.

6.8 Robustness in the two-disease variant

Two earlier analyses, run without retraining in the two-disease variant on the Rio surfaces, complement the cross-geography study. Across epidemic years (dengue 2015 with chikungunya 2016, and Rio 2016 alone), results are practically unchanged (C: +2692, +2710, +2721; C $\times$ A between +2009 and +2077, $P \geq 0.98$), as expected from surfaces that are all about 54% informative. Across the whole range of spatial informativeness obtained by sharpening, from 0.50 to 0.94, the advantage of per-case credit holds (C $\times$ A between +2000 and +2350, P between 0.89 and 1.00; Figure 12). C’s reward stays flat: an agent trained where location is uninformative does not exploit it when it becomes informative—consistent with the idealized-geography result, where only an agent *trained* there learns to use the map.

7 Discussion

The obstacle to learning a testing policy was not where it first seemed. For much of this project, “the agent does not test” was treated as a problem of exploration, reward scale or algorithm. Part

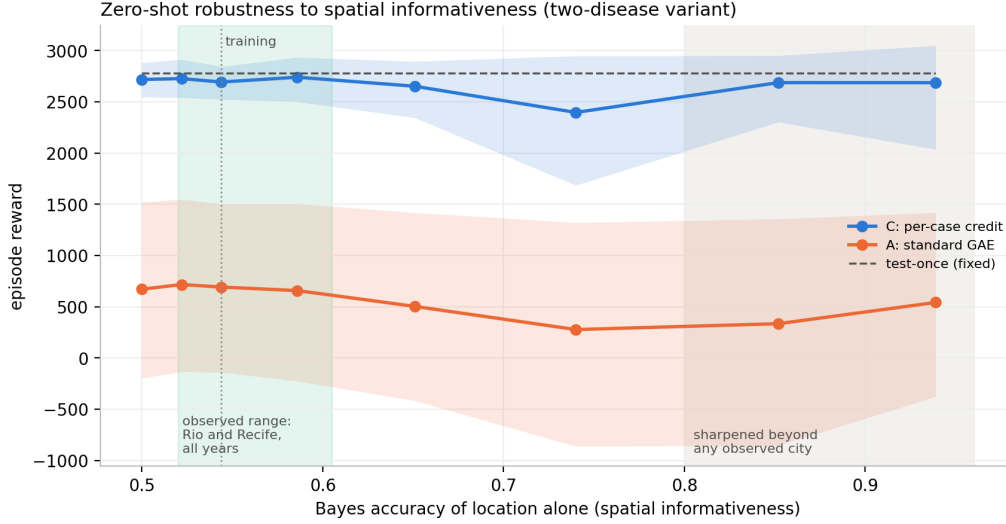


Figure 12: Zero-shot evaluation as spatial informativeness varies (two-disease variant; 95% intervals over training seeds and outbreaks). The green band is the range observed in Rio de Janeiro and Recife; the gray band requires sharpening beyond any observed city.

of it was: a stabilized DQN reliably learns inaction, and PPO was the first learned agent to beat doing nothing. But the decisive bottleneck was credit assignment. When decisions about many patients interleave in epidemic time, the standard return carries almost no information about whether a test was useful; the return along the patient’s own timeline carries almost all of it. Estimating advantages per case, in days, recovers that signal without touching the environment or the reward, and the result holds in every geography we tested.

What the agent learns depends on the problem it faces. With only two diseases, one test suffices and the agent learns *whether* to investigate. With non-arboviral suspects, investigation has depth, and the agent learns a policy as good as sequential testing guided by the clinician—the best rule we wrote a priori—while ordering 7% fewer tests. Opening the policy case by case shows that the saving has a simple source, trusting the clinician’s non-arboviral label, and that the rest of the policy *is* the sequential rule. Written as a rule, that refinement is as good as the expert rule with fewer tests, and better when tests are expensive. Where location is informative, the agent learns to use the map and surpasses every rule with far fewer tests. Rediscovering, and improving, the expert rule where a simple rule is near-optimal, and beating it where there is exploitable structure, is the pattern one would hope for from a policy learned from reward alone; the case-level analysis is what turns the first half of that statement from an aggregate tie into an explanation. A useful by-product is itself practical: among fixed rules, testing first for the disease the clinician suspects is substantially better than always testing dengue first (+350 in reward and 73 fewer tests per episode).

The transfer study turns the weak spatial signal of real data into a design lesson. Because location carries little information in real cities, the best policies there are spatially invariant: the Rio-trained agent ignores its spatial inputs and, for that reason, transfers to another city. Policies that do rely on space are brittle—the idealized-geography agent because the signal disappears in real cities, the first Recife agent because it found a cue that only our simulator provided. That episode is a caution for simulation-based policy learning: the agent exploited an artifact that we had not noticed, the aggregate evaluation could not tell a shortcut from poor learning, and it was input ablations and a cross-city test that located it. Once the artifact is removed, Rio de Janeiro and Recife are interchangeable as training grounds. Mixing very different geographies, on the other hand, did not help: one policy had to behave in opposite ways and learned neither. Making spatial inputs fully comparable across cities requires more than a common scale: case

counts must scale with each city’s population too, and doing so removes the Rio-to-Recife loss that the common scale had created.

7.1 Limitations

- **Training seeds.** Five training seeds per arm and geography still give the upper level of the hierarchical bootstrap limited resolution; intervals are honest about outbreaks and somewhat optimistic about training runs. The diagnostic experiments of the first round (random placement, physical scale) used three seeds.
- **Reuse of the test outbreaks.** The derived rule and the correction of the simulator were designed after looking at results on the first 30 test outbreaks. A fresh set confirmed the main conclusions and qualified two (Section 6.7); the sensitivity sweeps were not repeated on it.
- **Expensive tests.** Retraining at a test cost of 8 failed to beat inaction; the method was not shown to find the narrow policies that pay when tests are very expensive.
- **Spatial scale.** In the main experiments the grid is 400×400 for every city, so its physical scale differs between cities (the 20-cell radius of epidemiological confirmation is about 1 km in Recife and 3 km in Rio). A common physical scale alone creates a density shift, because the number of cases per episode does not scale with the city’s population; scaling the cases with the population removes the shift (Section 6.5), but only for two cities and one reference outbreak size, and the agent trained on the scaled (smaller) Recife outbreaks does not transfer to larger ones. These results were obtained on the first 30 test outbreaks and were not repeated on the fresh set.
- **Simulated clinicians and tests.** Clinician accuracy is symmetric between diseases and drawn per episode; test sensitivity does not depend on the day of collection; non-arboviral suspects are uniform over the inhabited area. Section 6.6 shows the conclusions are robust to the levels of these parameters, not to their structure. Recife’s discarded notifications could provide an empirical surface, but a notification discarded for dengue may be chikungunya.
- **Data.** Rio surfaces are built from notified cases and Recife surfaces from confirmed cases geocoded to street or neighborhood level. In 2015–2016 Recife was the epicenter of the Zika epidemic, and part of the dengue confirmed by clinical–epidemiological criteria in those years is likely Zika.
- **Externality of tests.** Per-case credit does not credit a test for its effect on future epidemiological confirmations nearby; the agents barely use epidemiological confirmation.
- **Capacity and seasonality.** Scarcity is modeled through test cost, not a daily laboratory capacity; the SEIR model has no seasonal forcing.

7.2 Future work

The most immediate step is additional years of Rio de Janeiro notifications as held-out tests. To make the choice of *which* cases to test twice a real decision, the simulator needs case-level heterogeneity: symptom profiles with disease-specific likelihood ratios, days since symptom onset (which govern RT-PCR sensitivity), and clinicians or health units of different, observable reliability. Beyond that: synthetic populations from the 2022 Brazilian census [49] to build scenarios for other cities with realistic population density; a daily laboratory capacity, which would turn scarcity into explicit prioritization; and a credit assignment that accounts for the externality of positive tests.

8 Conclusion

We framed the allocation of laboratory tests in arbovirus surveillance as a sequential decision in epidemic time, in a simulator grounded in surveillance data from two Brazilian cities. In real data, the location of a case barely reveals whether it is dengue or chikungunya, so tests are the main source of information—and standard deep RL fails to learn to use them, because the value of a test is lost among interleaved decisions about other patients. Estimating advantages along each patient’s timeline, in epidemic days, makes the agent investigate in every geography we tested. In real cities it matches the best expert rule we wrote with fewer tests, robustly to the simulated test and clinician parameters and on fresh outbreaks, and a case-by-case reading of its decisions turns what it learned into a frugal fixed rule; where location is informative, it beats every rule outright. Which rule is best depends on the case mix. Whether a learned policy transfers depends on whether it leans on spatial cues that the next city does not share—including cues that exist only because of the simulator, which ablations and cross-city tests can expose.

References

- [1] Ministério da Saúde do Brasil. Arboviroses. Website, . URL <https://www.gov.br/saude/pt-br/assuntos/saude-de-a-a-z/a/arboviroses>. Acessado em 2 de setembro de 2025.
- [2] Cíntia Pereira Donateli and Fernando Celso de Campos. VISUALIZAÇÃO DE DADOS DE VIGILÂNCIA DAS ARBOVIROSES URBANAS TRANSMITIDAS PELO Aedes Aegypti em Minas Gerais, Brasil. *JISTEM - Journal of Information Systems and Technology Management*, 20:e202320003, 2023. ISSN 1807-1775. doi: <https://doi.org/10.4301/S1807-1775202320003>. URL <https://www.scielo.br/j/jistem/a/VjDXD9bcKGsxXXVdSwNZ4kF/?lang=pt>. Publisher: TECSI Laboratório de Tecnologia e Sistemas de Informação - FEA/USP.
- [3] Ministério da Saúde do Brasil. Boletim epidemiológico: Monitoramento dos casos de arboviroses urbanas. Technical report, Secretaria de Vigilância em Saúde e Ambiente, 2024. Os números de volume e edição podem ser atualizados conforme a publicação mais recente consultada.
- [4] João Bosco Siqueira Junior, Eduardo Massad, Abner Lobao-Neto, Randee Kastner, Louisa Oliver, and Elaine Gallagher. Epidemiology and costs of dengue in Brazil: a systematic literature review. *International Journal of Infectious Diseases*, 122:521–528, September 2022. ISSN 1201-9712. doi: 10.1016/j.ijid.2022.06.050. URL <https://www.sciencedirect.com/science/article/pii/S1201971222003836>.
- [5] Ministério da Saúde do Brasil. Chikungunya. Website, . URL <https://www.gov.br/saude/pt-br/assuntos/saude-de-a-a-z/c/chikungunya>. Acessado em 2 de setembro de 2025.
- [6] Wilson Arrubla-Hoyos, Jorge Gómez Gómez, and Emiro De-La-Hoz-Franco. Methodology for the Differential Classification of Dengue and Chikungunya According to the PAHO 2022 Diagnostic Guide. *Viruses*, 16(7):1088, July 2024. ISSN 1999-4915. doi: 10.3390/v16071088.
- [7] Zheng Yu, Yikuan Li, Joseph Kim, Kaixuan Huang, Yuan Luo, and Mengdi Wang. Deep Reinforcement Learning for Cost-Effective Medical Diagnosis, February 2023. URL <http://arxiv.org/abs/2302.10261>. arXiv:2302.10261 [cs].
- [8] Reza Yaesoubi and Ted Cohen. Generalized Markov models of infectious disease spread: A novel framework for developing dynamic health policies. *European Journal of Operational Research*, 215(3):679–687, 2011.

- [9] Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction*. MIT press, 2018.
- [10] Gymnasium Documentation. URL <https://gymnasium.farama.org/index.html>.
- [11] Maxim Lapan. *Deep Reinforcement Learning Hands-On: Apply Modern RL Methods to Practical Problems of Chatbots, Robotics, Discrete Optimization, Web Automation, and More*. Packt Publishing, second edition, 2020. ISBN 978-1-83882-699-4.
- [12] Hamsa Bastani, Kimon Drakopoulos, Vishal Gupta, Ioannis Vlachogiannis, Christos Hadjichristodoulou, Pagona Lagiou, Gkikas Magiorkinis, Dimitrios Paraskevis, and Sotirios Tsiodras. Efficient and targeted COVID-19 border testing via reinforcement learning. *Nature*, 599:108–113, 2021. doi: 10.1038/s41586-021-04014-z.
- [13] Yu-Shao Peng, Kai-Fu Tang, Hsuan-Tien Lin, and Edward Chang. REFUEL: Exploring sparse features in deep reinforcement learning for fast disease diagnosis. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [14] Hajin Shim, Sung Ju Hwang, and Eunho Yang. Joint active feature acquisition and classification with variable-size set encoding. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [15] Stephen G. Pauker and Jerome P. Kassirer. The threshold approach to clinical decision making. *New England Journal of Medicine*, 302(20):1109–1117, 1980. doi: 10.1056/NEJM198005153022003.
- [16] Pieter J. K. Libin, Arno Moonens, Timothy Verstraeten, Fabian Perez-Sanjines, Niel Hens, Philippe Lemey, and Ann Nowé. Deep reinforcement learning for large-scale epidemic control. In *Machine Learning and Knowledge Discovery in Databases. Applied Data Science and Demo Track (ECML PKDD 2020)*, volume 12461 of *Lecture Notes in Computer Science*, pages 155–170. Springer, 2021. doi: 10.1007/978-3-030-67670-4_10.
- [17] Peter Sunehag, Guy Lever, Audrunas Gruslys, Wojciech Marian Czarnecki, Vinicius Zambaldi, Max Jaderberg, Marc Lanctot, Nicolas Sonnerat, Joel Z. Leibo, Karl Tuyls, and Thore Graepel. Value-decomposition networks for cooperative multi-agent learning based on team reward. In *Proceedings of the 17th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*, pages 2085–2087, 2018.
- [18] Tabish Rashid, Mikayel Samvelyan, Christian Schroeder de Witt, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *PMLR*, pages 4295–4304, 2018.
- [19] Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- [20] Cathy Wu, Aravind Rajeswaran, Yan Duan, Vikash Kumar, Alexandre M. Bayen, Sham Kakade, Igor Mordatch, and Pieter Abbeel. Variance reduction for policy gradient with action-dependent factorized baselines. In *International Conference on Learning Representations (ICLR)*, 2018.
- [21] Jose A. Arjona-Medina, Michael Gillhofer, Michael Widrich, Thomas Unterthiner, Johannes Brandstetter, and Sepp Hochreiter. RUDDER: Return decomposition for delayed rewards. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

- [22] Anna Harutyunyan, Will Dabney, Thomas Mesnard, Mohammad Gheshlaghi Azar, Bilal Piot, Nicolas Heess, Hado van Hasselt, Greg Wayne, Satinder Singh, Doina Precup, and Rémi Munos. Hindsight credit assignment. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [23] Steven J. Bradtke and Michael O. Duff. Reinforcement learning methods for continuous-time Markov decision problems. In *Advances in Neural Information Processing Systems*, volume 7, pages 393–400, 1995.
- [24] Karl Cobbe, Oleg Klimov, Chris Hesse, Taehoon Kim, and John Schulman. Quantifying generalization in reinforcement learning. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *PMLR*, pages 1282–1289, 2019.
- [25] Robert Kirk, Amy Zhang, Edward Grefenstette, and Tim Rocktäschel. A survey of zero-shot generalisation in deep reinforcement learning. *Journal of Artificial Intelligence Research*, 76: 201–264, 2023.
- [26] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020. doi: 10.1038/s42256-020-00257-z.
- [27] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017.
- [28] Peter J. Diggle, Jonathan A. Tawn, and R. A. Moyeed. Model-based geostatistics. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 47(3):299–350, 1998.
- [29] Jared Aldstadt, In-Kyu Yoon, Darunee Tannitisupawong, Richard G. Jarman, Stephen J. Thomas, Robert V. Gibbons, Angkana Uppapong, Sopon Iamsirithaworn, Alan L. Rothman, Thomas W. Scott, and Timothy Endy. Space-time analysis of hospitalised dengue patients in rural Thailand reveals important temporal intervals in the pattern of dengue virus transmission. *Tropical Medicine & International Health*, 17(9):1076–1085, 2012. doi: 10.1111/j.1365-3156.2012.03040.x.
- [30] Mark Towers, Jordan K Terry, Ariel Kwiatkowski, John U Balis, Gianluca Cola, Tristan Deleu, Manuel Goulão, Andreas Kallinteris, Arjun KG, Markus Krimmel, et al. Gymnasium: A standard interface for reinforcement learning environments, 2023.
- [31] Syafruddin Side and MSM Noorani. Sir-network model and its application to dengue fever. *World Journal of Modelling and Simulation*, 11(2):106–113, 2015.
- [32] M Derouich, A Boutayeb, and EH Twizell. A model of dengue fever. *BioMedical Engineering OnLine*, 2:4, February 2003. ISSN 1475-925X. doi: 10.1186/1475-925X-2-4. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC153427/>.
- [33] Miranda Chan and Michael A. Johansson. The incubation periods of dengue viruses. *PLoS ONE*, 7(11):e50972, 2012. doi: 10.1371/journal.pone.0050972.
- [34] Lauren B. Carrington and Cameron P. Simmons. Human to mosquito transmission of dengue viruses. *Frontiers in Immunology*, 5:290, 2014. doi: 10.3389/fimmu.2014.00290.
- [35] Daniel A. M. Villela, Leonardo S. Bastos, Lucas M. de Carvalho, Oswaldo G. Cruz, Marcelo F. C. Gomes, Betina Durovni, Maria C. Lemos, Valeria Saraceni, Flavio C. Coelho, and Claudia T. Codeço. Zika in Rio de Janeiro: Assessment of basic reproduction number and

- comparison with dengue outbreaks. *Epidemiology and Infection*, 145(8):1649–1657, 2017. doi: 10.1017/S0950268817000358.
- [36] Filipe Romero Rebello Moreira, Mariane Talon de Menezes, Clarisse Salgado-Benvindo, Charles Whittaker, Victoria Cox, Nilani Chandradeva, Hury Hellen Souza de Paula, André Frederico Martins, Raphael Rangel das Chagas, Rodrigo Decembrino Vargas Brasil, Darlan da Silva Cândido, Alice Laschuk Herlinger, Marisa de Oliveira Ribeiro, Monica Barcellos Arruda, Patricia Alvarez, Marcelo Calado de Paula Tôrres, Ilaria Dorigatti, Oliver Brady, Carolina Moreira Voloch, Amilcar Tanuri, Felipe Iani, William Marciel de Souza, Sergian Vianna Cardozo, Nuno Rodrigues Faria, and Renato Santana Aguiar. Epidemiological and genomic investigation of chikungunya virus in Rio de Janeiro state, Brazil, between 2015 and 2018. *PLoS Neglected Tropical Diseases*, 17(9):e0011536, 2023. doi: 10.1371/journal.pntd.0011536.
 - [37] Prefeitura do Recife. Casos de dengue, zika e chikungunya. Portal de Dados Abertos da Cidade do Recife. Open Database License (ODbL), 2026. URL <https://dados.recife.pe.gov.br/dataset/casos-de-dengue-zika-e-chikungunya>. Accessed 24 Sep 2026.
 - [38] Prefeitura do Recife. Área urbana: bairros e trechos de logradouros. Portal de Dados Abertos da Cidade do Recife, 2026. URL <https://dados.recife.pe.gov.br/dataset/area-urbana>. Accessed 24 Sep 2026.
 - [39] Gilberto A. Santiago, Edgardo Vergne, Yashira Quiles, Joan Cosme, Jesus Vazquez, Juan F. Medina, Freddy Medina, Candimar Colón, Harold Margolis, and Jorge L. Muñoz-Jordán. Analytical and clinical performance of the CDC real time RT-PCR assay for detection and typing of dengue virus. *PLoS Neglected Tropical Diseases*, 7(7):e2311, 2013. doi: 10.1371/journal.pntd.0002311.
 - [40] Jesse J. Waggoner, Lionel Gresh, Alisha Mohamed-Hadley, Gabriela Ballesteros, Maria Jose Vargas Davila, Yolanda Tellez, Malaya K. Sahoo, Angel Balmaseda, Eva Harris, and Benjamin A. Pinsky. Single-reaction multiplex reverse transcription PCR for detection of Zika, chikungunya, and dengue viruses. *Emerging Infectious Diseases*, 22(7):1295–1297, 2016. doi: 10.3201/eid2207.160326.
 - [41] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.
 - [42] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017. URL <https://arxiv.org/abs/1707.06347>.
 - [43] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016. URL <https://arxiv.org/abs/1506.02438>.
 - [44] Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the Sixteenth International Conference on Machine Learning (ICML)*, pages 278–287, 1999.
 - [45] Timothy C. Frommeyer, Michael M. Gilbert, Reid M. Fursmidt, Youngjun Park, John Paul Khouzam, Garrett V. Brittain, Daniel P. Frommeyer, Ean S. Bett, and Trevor J. Bihl. Reinforcement Learning and Its Clinical Applications Within Healthcare: A Systematic Review of Precision Medicine and Dynamic Treatment Regimes. *Healthcare*, 13(14):1752, July 2025. ISSN 2227-9032. doi: 10.3390/healthcare13141752. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC12295150/>.

- [46] Jiayi Weng, Huoran Chen, Dong Yan, Kaichao You, Alexis Duburcq, Ming der Wang, Hang Su, and Jun Zhu. Tianshou: A highly modularized deep reinforcement learning library. *Journal of Machine Learning Research*, 22(255):1–6, 2021. URL <http://jmlr.org/papers/v22/21-0883.html>.
- [47] Varun Saravanan, Gordon J. Berman, and Samuel J. Sober. Application of the hierarchical bootstrap to multi-level data in neuroscience. *Neurons, Behavior, Data analysis, and Theory*, 3(5):1–25, 2020.
- [48] Rishabh Agarwal, Max Schwarzer, Pablo Samuel Castro, Aaron Courville, and Marc G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL <https://arxiv.org/abs/2108.13264>.
- [49] Mosqlimate Project. synthetic-pop: população sintética a partir do Censo IBGE 2022 e do CNEFE. Repositório GitHub. URL <https://github.com/Mosqlimate-project/synthetic-pop>. Acesso em: set. 2026.

A Work in progress (internal; remove before submission)

- **[TODO: Held-out years of Rio de Janeiro (data requested from Fiocruz) as additional test geographies; calibrate clinician accuracy, test sensitivity by day of collection and symptoms with the Recife microdata.]**